A-A-53132
17 August 1987

COMMERCIAL ITEM DESCRIPTION

WHEELS, ABRASIVE (Aluminum Oxide, Straight Handpiece)

The General Services Administration has authorized the use of this Commercial Item Description.

This commercial item description covers mandrel mounted straight handpiece aluminum oxide wheels.

The wheels shall be of the following shapes specified herein:

Shape 1 - Square edge

Shape 2 - Round edge

Salient characteristics:

The wheels shall be suitable for use in sharpening delicate cutting edges of periodontal instruments. Shape 1 shall be a square edge mounted wheel. Shape 2 shall be a round cone shape mounted wheel. Wheels shall be furnished in the shape for the applicable stock number. Shape and dimensions of the wheels shall be as shown in figure 1.

Material.

Abrasive and binder. The abrasive shall be 120 grit, high quality, friable aluminum oxide. The grain size shall conform to ANSI B74.12-76R2. The binder (bond) shall be high quality vitrified clay bond. The hardness and grain spacing shall be grade "M" and structure "No. 5". The abrasive and ceramic bond shall be of proper proportions and mix, and fired to form a grinding wheel suitable for sharpening dental cutting instruments. The wheels shall be securely mounted on the mandrel with high quality phenolic resin.

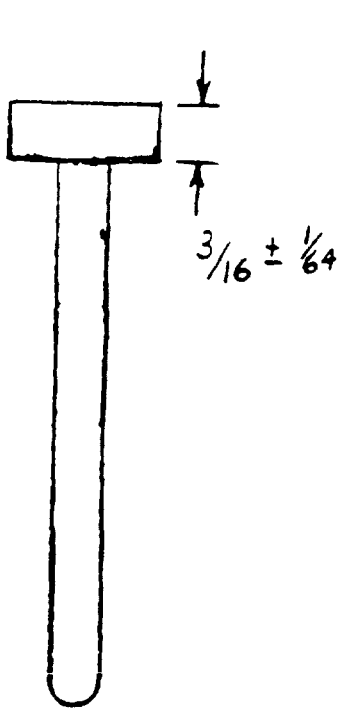
Mandrels. The mandrel shall be made of either suitably-plated carbon steel or passivated corrosion-resisting steel.

Hardness. The mandrel shall have a Rockwell hardness of C20 minimum, when tested as specified herein.

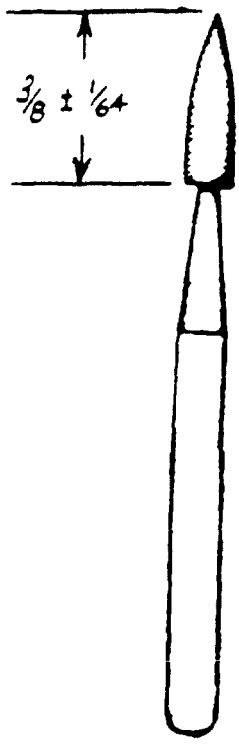
AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:15Z
Check the source to verify that this is the current version before use.



SHAPE 1



SHAPE 2

ALL DIMENSIONS ARE IN INCHES

	Diameter of head (± 0.015)	Diameter of mandrel shank	Overall length (± 1/16)
SHAPE 1	1/2	0.0919 - 0.0925	1-3/4
SHAPE 2	1/4*	0.0919 - 0.0925	1-7/8

* 10 ± 2 degree taper

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:15Z
Check the source to verify that this is the current version before use.

Runout. The runout shall not be more than 0.006 inch overall, when tested as specified herein.

Finish. The mandrel shall have a bright smooth finish. The end of mandrel opposite the concealed portion shall be rounded.

Performance. The wheels shall produce a surface finish having maximum roughness value of 10 micro inches when tested as specified herein.

Workmanship. The mounted wheels shall be free from defects which detract from their appearance and may impair their serviceability.

Unit. Package (PG). One package containing six wheels and detailed, instructions for use, as specified constitute one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time or from time to time during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item. Inspection shall be conducted to determine compliance with specification requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:15Z
Check the source to verify that this is the current version before use.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be six wheels. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, and sampling for tests (when applicable) shall be conducted at inspection level S-1, AQL 1.0.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the Contracting Officer to determine if the product is acceptable.

The Contracting Officer has the option of accepting or rejecting the product.

Tests.

Abrasive particle size. The determination of grain size shall be in accordance with the procedures set forth in ANSI B74.12-76R2.

Hardness. Hardness shall be determined in accordance with method 243 (Rockwell) or method 244 (Diamond Pyramid Hardness) of FED-STD-151.

Runout. The shank of the mandrel shall be rotated in a vee block by means of a round file; the runout at the neck of the mandrel shall be measured by means of a dial gage. The total movement shown on the gage shall be the overall runout.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:15Z
Check the source to verify that this is the current version before use.

Performance. The wheels shall be used for honing an AISI 440A corrosion-resisting steel, 7/32 inch octagonal bar stock with a Rockwell hardness of C50 to C60. The surface roughness shall be determined by a Brush Surfindicator or other suitable equipment. Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

NSN	ITEM IDENTIFICATION
6520-00-787-2888	WHEEL, ABRASIVE Aluminum Oxide, Straight Handpiece, Square Edge, 1/2 by 1/4 inches, No. 303, 6s.
6520-00-787-2890	WHEEL ABRASIVE, Aluminum Oxide, Straight Handpiece, Rounded Cone, 1/4 by 3/8 inches, No. 300, 6s.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:15Z
Check the source to verify that this is the current version before use.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

05445

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

Project No. 6520-2011

Location: NCR CID DISK 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:15Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge (*DO NOT STAPLE*), and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

Fold along this line)

DEFENSE LOGISTICS AGENCY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 4966 WASHINGTON D.C.

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Personnel Support Center
ATTN: DPSC-ATTS
2800 South 20th Street
Philadelphia, Pennsylvania 19101-8419

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:15Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-53132	WHEELS, ABRASIVE, ALUMINUM OXIDE, STRAIGHY HANDPIECE

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street City State ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last First MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:15Z
Check the source to verify that this is the current version before use.